* Assignment No:- 4 *

Q.1]

```
* * * * *
* * * * *
* * * * *
```

→ i = 3
j = 5

```
for (int i = 1 ; i ≤ 3 ; i++) {
        for (int j = 1 ; j ≤ 5 ; j++) {
                sop ("* ");
        }
        sopln ();
}
```

→ Time Complexity = [$O(n^2)$]
Space Complexity = $O(1)$

Q.2]

```
* * * *
* * * *
* * * *
* * * *
```

→ i = 4 j = 4

```
int n = 4 ;
  for (int i = 1 ; i ≤ n ; i++) {
        for (int j = 1 ; j ≤ n ; j++) {
                sop ("*" + " _ ");
        }
        sopln ();
}
```

TC = $O(n^2)$
SC = $O(1)$

Q.3]

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

i = 4 j = 4

```
int n=4;
for(int i=1; i≤n; i++){
    for(int j=1; j≤n; j++){
        sop(j + "_");
    }
    sopln();
}
```

TC = $O(n^2)$
SC = $O(1)$

Q.4]

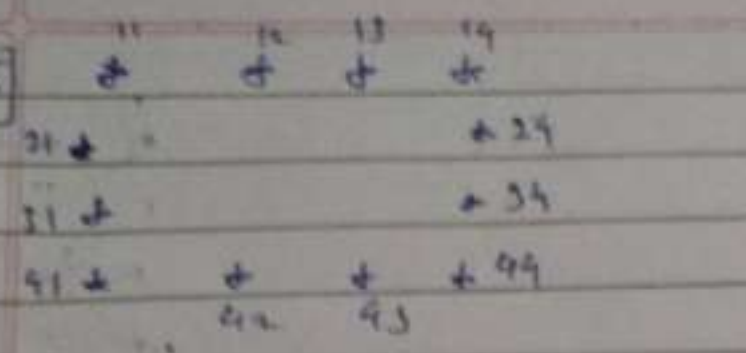
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

i = 4
j = 4

```
int n=4;
for(int i=1; i≤n; i++){
    for(int j=1; j≤n; j++){
        sop(i + "_");
    }
    sopln();
}
```

TC = $O(n^2)$
SC = $O(1)$

Q5]



$i = 4$

$j = 4$

if $j == 1 \mid\mid j == 4$

$j == 1 \mid\mid j == 4$

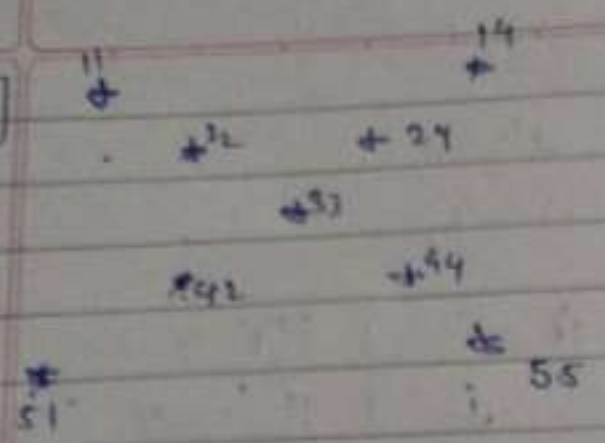
```
int n = 4;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {

        if ( j == 1 || j == 4 || j == 1 || j == 4) {
            cop ("* ");
        }
        else {
            cop ("  ");
        }
    }
    copln();
}
```

$TC = O(n^2)$

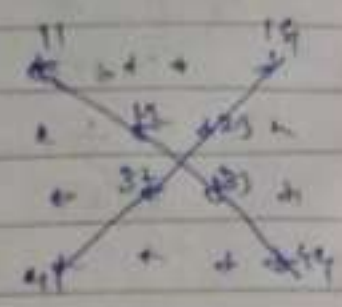
$SC = O(1)$

Q.8]



$i = 5$

$j = 5$



$n+1$

```
int n=5;

for(int i=1; i<=n; i++){
    for (int j=1; j<n; j++){
        if( i==j || i+j==n+1){
            sop("* ");
        }
        else {
            sop("  ");
        }
    }
    sopln();
}
```

$TC = O(n^2)$

$SC = O(1)$

Q.7]

```
1 2 3
4 5 6
7 8 9
```

i = 3
j = 3

```
int n = 3;
int count = 1;

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        sop (count + "_ ");
        count ++;
    }
    sopln();
}
```

TC = O(n²)
SC = O(1)

Q.8)

```
*
* *
* * *
* * * *
* * * * *
```

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        sop ("*" + "_ ");
    }
    sopln();
}
```

TC = O(n²)
SC = O(1)

Q8]

```
* * * * *
* * * *
* * *
* *
*
```

```
int n=5

    for(int i= n; i<=1 ; i--){
        for(int j=1 ; j<= i; j++){
            sop("*"+"-");
        }
        sopln();
    }
```

TC = $O(n^2)$
SC = $O(1)$

Q9]

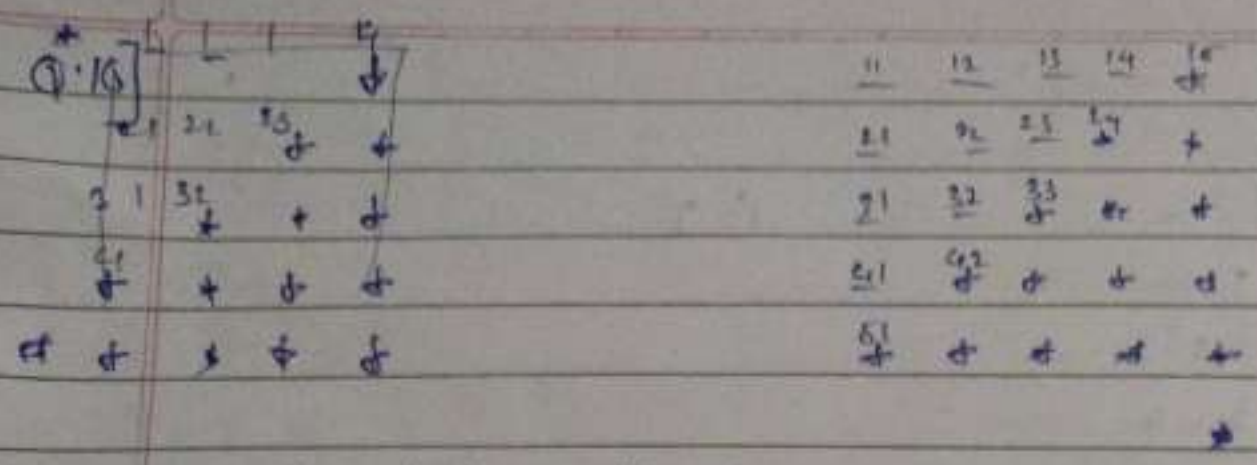
```
1  2  3  4
*  #  *  #
2  3  4  5
#  *  #  *
3  4  5  6
*  #  *  #
4  5  6  7
#  *  #  *
```

i = 4
j = 4

```
int n=4
for(int i=1 ; i<=n; i++){
    for(int j=1 ; j<=n; j++){
        if(i+j %2 == 0){
            sop("* ");
        }
        else {
            sop("# ");
        }
    }
    sopln();
}
```

TC = $O(n^2)$
SC = $O(1)$

Q.10



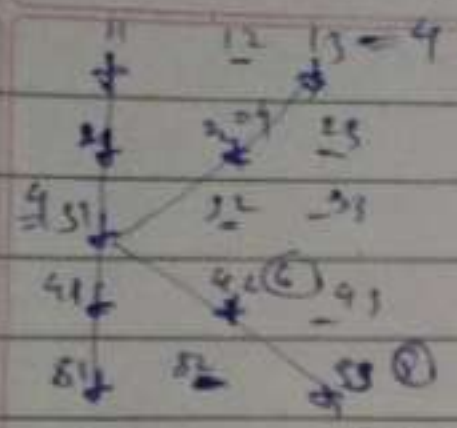
6 and greater than 6.

```
int n = 5;
for (int i = 1; i ≤ n; i++){
    For (int j = 1; j ≤ n; j++) {
        if (i+j ≥ n+1){
            sop("* ");
        }
        else {
            sop("_ _");
        }
    }
    sopln();
}
```

TC = $O(n^2)$

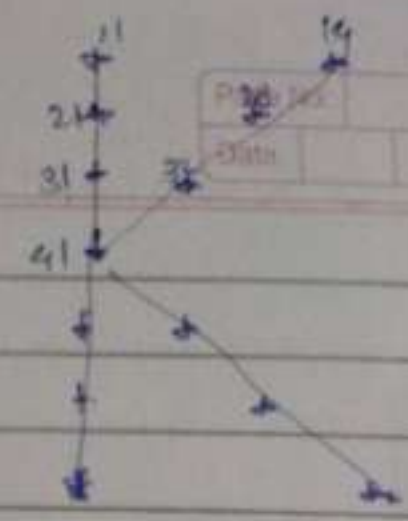
SC = $O(1)$

Q.12]



$S = 3$

$i - r$



```
int n=3; K=5
    for(int i=1; i<=K; i++){
        for(int j=1; j<=n; j++){
            if( j==1 || i+j==n+1 ||
                (i+j)%2==0 && i>j ){
                Sop("* ");
            }
            else{
                SOP("  ");
            }
        }
        Sopln();
    }
```

TC = O(n*k)

SC = O(1)